American Computer Science League

2023-2024 ● Contest 1: Short Problems ● Classroom Division

1. Computer Number Systems

Convert 2324_{10} to octal.

A. 4442
B. 4244
C. 2444
D. 4424
E. 4224

2. Computer Number Systems

What is the value of $73452_8 - 4261_8 + 752_8$ in base 16?

A. 27771
B. 70143
C. 7063
D. 70163
E. 7043

3. Recursive Functions

Find $f(22)$ given:

$$f(x) = \begin{cases} f(x-4)+1 & \text{if } x \geq 12 \\ 2 \cdot f(x+1) - 3 & \text{if } 8 < x < 12 \\ x - 3 & \text{if } x \leq 8 \end{cases}$$

A. 3
B. 6
C. 9
D. 17
E. 18

4. Recursive Functions

Find $f(f(24))$ given:

$$f(x) = \begin{cases} f\left(\frac{x}{2}\right)+4 & \text{if } x \text{ is even} \\ f(x-1)+2 & \text{if } x \text{ is odd and prime} \\ x & \text{otherwise} \end{cases}$$

Remember that 1 is not a prime number.

A. 21
B. 19
C. 15
D. 11
E. 9

5. What Does This Program Do? (Branching)

Sales employees are often paid based on commission. Using the following conditions:

sales < 5000, 1% of sales
5000 ≤ sales < 10000, 1% of the first $5,000 and 3% of the rest
sales ≥ 10000, 1% of the first $5,000, 3% of the next $5,000, and 5% of the amount over $10,000

List all of the lines of code (a,b,c,d) that will correctly calculate a salesperson's commission (stored in variable `cpay`) for any positive value of sales (given in variable `sales`)?

a.
```
if sales >= 10000 then
    cpay = sales*.05 + 5000*.03 + 5000*.01
else if sales >= 5000 then
    cpay = sales*.03 + 5000*.01
else
    cpay = sales*.01
end if
```

b.
```
if sales < 5000 then
    cpay = sales*.01
else if sales < 10000 then
    cpay = (sales-5000)*.03 + 5000*.01
else
    cpay = (sales-10000)*.05 + 5000*.03 + 50
end if
```

c.
```
if sales < 5000  then
    cpay = sales*.01
end if
if sales >= 5000 && sales < 10000 then
    cpay = (sales-5000)*.03 + 5000*.01
else if sales >= 10000 then
    cpay = (sales-10000)*.05 + 5000*.03 + 50
end if
```

d.
```
if sales >= 10000 then
    cpay = (sales-10000)*.05 + 5000*.03 + 50
else if sales >= 5000 then
    cpay = (sales-5000)*.03 + 5000*.01
else
   cpay = sales*.01
end if
```

A. b, c, d
B. a, c, d
C. a, b, c
D. a, b, d
E. a, b, c, d

6. Computer Number Systems

A palindrome is a word, phrase, or number that is the same forwards and backwards. For example, *madam* and *12321* are palindromes.

What is the next year in our calendar when its octal representation is a palindrome?

A. 2032
B. 3773
C. 4004
D. 2043
E. 2112

7. Computer Number Systems

Convert the following hex number to octal. What octal digit occurs most often?

$F43A6C3091_{16}$

A. 0
B. 1
C. 2
D. 3
E. 7

8. Recursive Functions

Find $f(17, 11)$ given:

$$f(x,y) = \begin{cases} f(x-1, y-2) + 3 & \text{if } x < y \\ f(y-1, x-2) + 1 & \text{if } x > y \\ x - y & \text{if } x = y \end{cases}$$

A. 8
B. 9
C. 12
D. 15
E. 16

9. Recursive Functions

Find $f(24)$ given:

$$f(x) = \begin{cases} 2 \cdot f(x-4) & \text{if } x \geq 18 \\ f(x+1) - 1 & \text{if } 15 < x < 18 \\ x + 2 & \text{if } x \leq 15 \end{cases}$$

A. 16
B. 32
C. 60
D. 120
E. 240

10. What Does This Program Do?

What is the output of the following program?

```
for yr = 2021 to 2025
    num = yr
    s1 = 0
    s2 = 0
    while num != 0
        d = num % 10
        s1 = s1 + d
        s2 = s2 + d^2
        num = int(num/10)
    end while
    if yr%s1 == 0 && yr%s2 == 0 then
        output yr
    end if
next yr
```

A. 2021
B. 2022
C. 2023
D. 2024
E. 2025